

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-7005(4)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Life Cycle Costing of Structures

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw flow charts and figures.

1. a) What does life cycle cost mean? What is the purpose of a life cycle cost estimate? 7
b) What are the various building-related costs? Explain each of them in brief. 7
2. a) If annual resource cost are $1/20$ of capital costs, by implementing measures that result in savings in resource cost of 15%. What will be the potential saving in the rent? 7
b) How LCC is a fundamental part of a decision-making process? Explain in detail. 7
3. a) What do you mean by Nett Present Value? Also explain its parameters. 7
b) Explain uncertainty and risk in detail. 7

CE-7005(4)-CBGS

PTO

[2]

4. You are evaluating whether to replace electric motors at a waste water treatment plant. Using an inflation rate of 3.5% and a salvage value of 15% of the initial costs. Determine which of these pumps is the most economical from a life cycle perspective. 14

Input	Option A	Option B	Option C	Option D
Cost	5,000/-	2,250/-	21,500/-	0
Annual Energy cost	11,000/-	6,700/-	5,500/-	11,000/-
Annual Maintenance	500/-	500/-	1,000/-	500/-
Annual Inspection	2,500/-	2,500/-	2,500/-	2,500/-
Life	8	6	12	5

5. a) Write short note on Break Even Analysis. 7
 b) Enlist the various LCCA-related software programs. How these softwares are proving their worth in LCCA? 7
6. What are the LCCA guidelines for Office of Management and Budget (OMB)? 14
7. Determine the equal annual end of year payment required over the life of these loans to repay them fully during the stated term. 14

Loan	Principal	Interest	Term (years)
A	1,20,000	8%	3
B	6,00,000	12%	10
C	75,000	10%	2
D	2,40,000	15%	5

8. Write short notes on: 14
- Break Even Point
 - Treatment of Inflation
 - Cash Flow

CE-7005(4)-CBGS